8.1 Answers: fractions, decimals and percentages

← Unit overview · Practice

For the parent: if an answer differs, ask “How did you get that?” before saying anything else. Often the method is right and one small step slipped. Praise the method you hear.

Set A: One number, three outfits

	Fraction	Decimal	Percentage	Strategy
1	¼	0,25	25%	Half of a half; a quarter of €1 is 25 cents.
2	¾	0,75	75%	Three quarters: 25 + 25 + 25.
3	1/10	0,1	10%	One tenth; 10 cents of a euro.
4	1/5	0,2	20%	0,2 = 2/10 = 1/5.
5	3/10	0,3	30%	30 out of 100 = 3 out of 10.
6	2/5	0,4	40%	1/5 = 20%, so 2/5 = 40%.
7	1/100	0,01	1%	One cent of a euro.
8	3/5	0,6	60%	0,6 = 6/10 = 3/5.
9	1/8	0,125	12,5%	Half of ¼: half of 25% is 12,5%.
10	½	0,5	50%	The first anchor.

Set B: Which is bigger?

1. **0,5.** ¼ = 0,25, which is less than 0,50.
2. **30%.** ¼ = 25%.
3. **¾.** ¾ = 0,75, more than 0,70.
4. **45%.** 2/5 = 40%.
5. **0,35.** 1/3 is 0,333…, a bit less than 0,35.
6. **½, 55%, 0,6.** As percentages: 50%, 55%, 60%.
7. **35%, 3/8, 0,4.** As decimals: 0,35, 0,375, 0,4.
8. **48%.** It is only 2% away from 50%; 0,45 is 5% away, 5/8 = 62,5% is 12,5% away.
9. **Noor.** Bram: 7/10 = 70%. Noor: 15/20 = 75%.
10. **Same.** 0,50 is just 0,5 with an extra zero.

Set C: Percentage of an amount

1. **€20.** Halve €40.
2. **€8.** 10% = divide by 10.
3. **50.** 25% = a quarter; 200 : 4.
4. **€3,50.** €35 : 10.
5. **€3.** 10% = €6, half of that = €3.
6. **€9.** 10% = €4,50, double = €9.
7. **90.** 25% = 30, so 75% = 3 × 30.
8. **6, then 18.** 1% = 600 : 100 = 6; 3% = 3 × 6.
9. **€12.** 10% = €8, 5% = €4, together €12.
10. **75 children.** 10% = 25, so 30% = 75.
11. **€9,50.** Half of €19.

Set D: Korting and btw

1. **€30.** Discount €30; €60 − €30.

2. **€30.** 25% of €40 = €10; €40 − €10.
3. **€72.** 10% = €8; €80 − €8.
4. **€200.** 10% = €25, 20% = €50; €250 − €50.
5. **20% korting.** Saved €9 out of €45; €9 is 1/5 of €45, and 1/5 = 20%.
6. **€25,50.** 10% = €3, 5% = €1,50, so 15% = €4,50; €30 − €4,50.
7. **€49.** 10% = €7, 30% = €21; €70 − €21.
8. **€121.** 21% of €100 = €21; €100 + €21.
9. **€242.** 10% = €20, 20% = €40, 1% = €2, so 21% = €42; €200 + €42.
10. **€2,18.** 9% of €2,00 is 18 cents (1% = 2 cents, × 9).
11. **€5.** Half of €10.

Set E: How many percent? Up and down

1. **75%.** 3/4 = 75%.
2. **24%.** 12/50 = 24/100.
3. **€22.** 10% of €20 = €2, added on.
4. **€6.** 20% of €5 = €1 more.
5. **20%.** 5 more out of the original 25 is 1/5 = 20%.
6. **25%.** 80 cents more out of the original €3,20; 80 is ¼ of 320.
7. **4 200.** 5% of 4 000 = 200, added on.
8. **25%.** 10 cm more out of the original 40 cm is ¼.
9. **75%.** 18/24 simplifies to 3/4.
10. **25% korting.** Saved €20 out of the original €80 = ¼.
11. **50%.** One half.

Puzzle corner

1. **No, it is 19%.** €100 → €90 after the first 10%. The second 10% is of €90, so €9 off → €81. You saved €19 out of €100.
2. **No.** €100 + 50% = €150. Then 50% of €150 is €75, so €150 − €75 = €75. The second percentage is taken from a bigger amount.
3. **Yes.** For example: a plant was 20 cm and is now 30 cm, so it is 150% of its old height. Or a school has 150% as many pupils as ten years ago.